

SOP and POS

CENG 2112.01 Lab for Digital Circuits

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Feb 17, 2026

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Abstract

An AND gate is a device that requires two inputs based off two options which are 0 and 1, then it will produce one output. If the inputs of this gate are the same (ex. Both inputs are 1), then the output will be the same as the inputs. Also, if one of the inputs are 0, then it will produce a 0. An OR gate is a device that requires two inputs based off two options similar to the AND gate and follows the same rules as far as the output produced will be the same as two identical inputs, but they are opposite in the sense that if one if the inputs is a 1, then it will produce a 1. An inverter will output the opposite input given to it, for example if a 1 is inputted into an inverter, then it will output a 0 and vise versa.

The important finding in this experiment is when you have a long sum of products, you can simplify it to a more simple circuit with the same output as before, then if you inverse it, you will get the product of sums version of it which will also have the same outputs.

Introduction

Similar to De Morgan's Theorem, the sum of products and product of sums of an expression will have equal outputs. To optimally use them is dependent on the output, when a few input combinations output 1, then it is best to use SOP, whereas POS is best when few input combinations output a 0. The purpose of these techniques is to simplify circuits based on truth tables.

Requirements

The requirement for this lab is to simulate the SOP of an equation then converting the SOP to a POS to show that their outputs are the same in Multisim with both systems receiving the same inputs. Then we used a logic analyzer within Multisim to check the waveforms of different outputs (1 and 0). Then we created these circuits on a bread board to produce and confirm our simulation is correct on Multisim. We also coded a schematic in Vivado using VHDL as well as to have a testbench of the waveforms to match the results from Multisim

Prelab

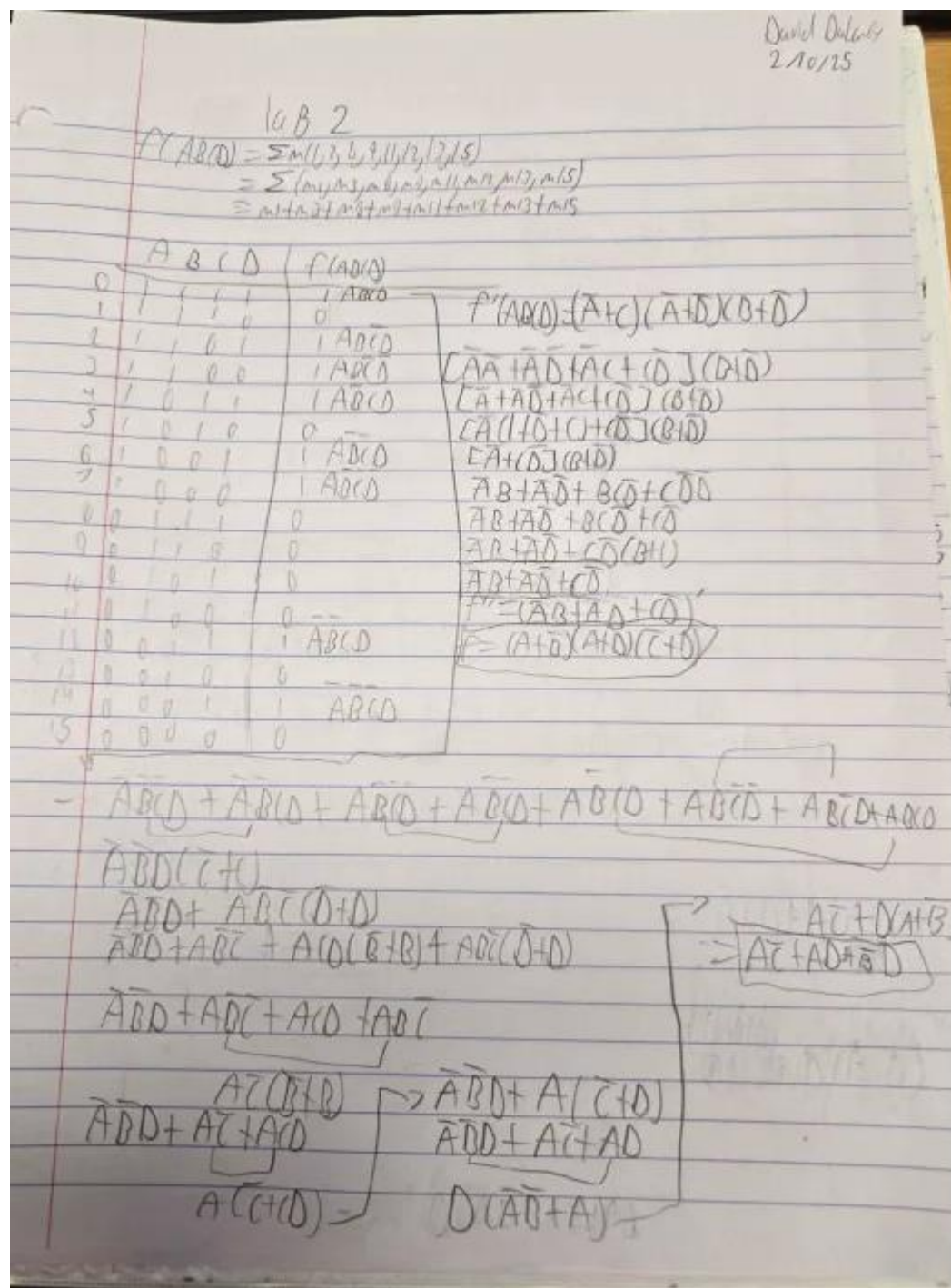


figure 1: prelab for SOP and POS of an expression with simplified expressions (boxed)

Implementation

The first step in implementation was to build a schematic with a word generator and a logic analyzer with a SOP circuit and a POS version of the SOP circuit based on five combinations of 0s and 1s to produce the same outputs. Then we simulated these combinations in VHDL to see if we get the same outcome as our schematics.

Design Code/Design Diagrams

For simplified equation:

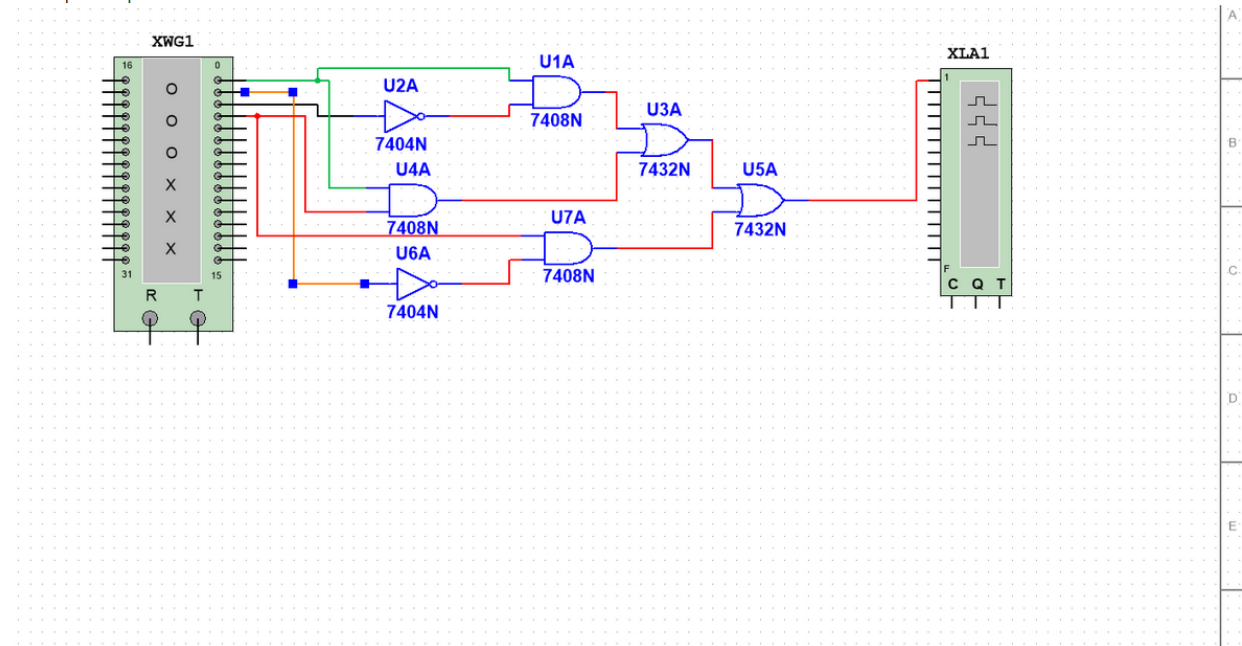


Figure 1: SOP circuit of the expression from the prelab

MULTISIM FOR T:

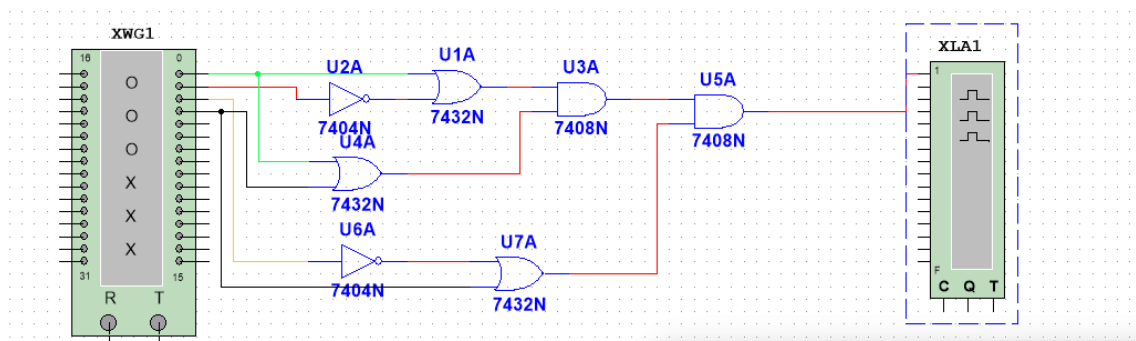


Figure 2: POS circuit of the SOP circuit

We concluded with these designs because they are the simplified version of the expression we worked on in the prelab. The first schematic is the circuit for the bottom boxed expression while the second schematic is the circuit for the top boxed expression. Similar to De Morgan's

theorem on an expression, the SOP and POS of an expression will produce the same outputs given the same inputs. This is shown in figures 6.1-6.10.

```
-----  
-- Company:  
-- Engineer:  
--  
-- Create Date: 02/10/2026 03:22:27 PM  
-- Design Name:  
-- Module Name: Lab2 - Behavioral  
-- Project Name:  
-- Target Devices:  
-- Tool Versions:  
-- Description:  
--  
-- Dependencies:  
--  
-- Revision:  
-- Revision 0.01 - File Created  
-- Additional Comments:  
--
```

```
-----  
  
library IEEE;  
use IEEE.STD_LOGIC_1164.ALL;
```

```
-- Uncomment the following library declaration if using
```

```

-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity Lab2 is
--creating ports for input and output
    Port ( A : in STD_LOGIC;
           B : in STD_LOGIC;
           C : in STD_LOGIC;
           D : in STD_LOGIC;
           F1 : out STD_LOGIC;
           F2 : out STD_LOGIC);
end Lab2;

architecture Behavioral of Lab2 is

begin
-- Inserted logic inside box
F1 <=((A and not(c))or((A and D)) or( not(B) and D));
F2 <=((A or not(B)) and(A or D)) and(C or D) ;

end Behavioral;

```

Schematics

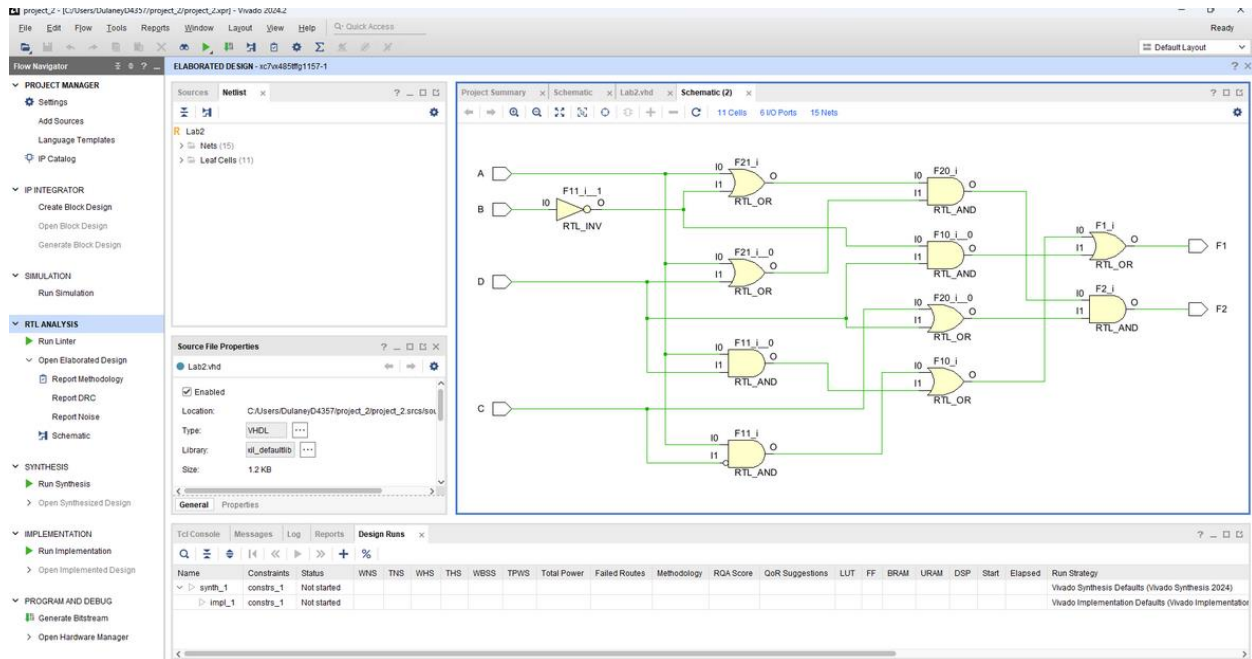


Figure 3: Schematic of the SOP circuit of the expression in the prelab along with the POS circuit

Testbench

-- Company:

-- Engineer:

--

-- Create Date: 02/10/2026 03:39:14 PM

-- Design Name:

-- Module Name: TB - Behavioral

-- Project Name:

-- Target Devices:

-- Tool Versions:

-- Description:

--

-- Dependencies:

--

-- Revision:

-- Revision 0.01 - File Created

-- Additional Comments:

--

library IEEE;

use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using

-- arithmetic functions with Signed or Unsigned values

--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating

-- any Xilinx leaf cells in this code.

--library UNISIM;

--use UNISIM.VComponents.all;

entity TB is

-- Port ();

end TB;

architecture Behavioral of TB is

--Place chip on board

Component DM is

```
Port ( A : in STD_LOGIC;
      B : in STD_LOGIC;
      C : in STD_LOGIC;
      D : in STD_LOGIC;
      F1 : out STD_LOGIC;
      F2 : out STD_LOGIC);
```

end Component;

--creating wires for each port.

signal WA, WB, WC, WD, WF1, WF2: std_logic; --instantiating wires for each port.

Begin

--Mapping each port to a wire..

MappingWires2BoxPorts: DM port map(A => WA,B=> WB,C => WC, D => WD, F1 => WF1, F2 => WF2);

process

begin

--Creating each input combination for four inputs

WA <= '1'; WB <= '1'; WC <= '1'; WD <= '1'; wait for 10ns;

WA <= '1'; WB <= '1'; WC <= '1'; WD <= '0'; wait for 10ns;

WA <= '1'; WB <= '1'; WC <= '0'; WD <= '1'; wait for 10ns;

WA <= '1'; WB <= '1'; WC <= '0'; WD <= '0'; wait for 10ns;

WA <= '1'; WB <= '0'; WC <= '1'; WD <= '1'; wait for 10ns;

WA <= '1'; WB <= '0'; WC <= '1'; WD <= '0'; wait for 10ns;

```

WA <= '1'; WB <= '0'; WC <= '0'; WD <= '1'; wait for 10ns;
WA <= '1'; WB <= '0'; WC <= '0'; WD <= '0'; wait for 10ns;
WA <= '0'; WB <= '1'; WC <= '1'; WD <= '1'; wait for 10ns;
WA <= '0'; WB <= '1'; WC <= '1'; WD <= '0'; wait for 10ns;
WA <= '0'; WB <= '1'; WC <= '0'; WD <= '1'; wait for 10ns;
WA <= '0'; WB <= '1'; WC <= '0'; WD <= '0'; wait for 10ns;
WA <= '0'; WB <= '0'; WC <= '1'; WD <= '1'; wait for 10ns;
WA <= '0'; WB <= '0'; WC <= '1'; WD <= '0'; wait for 10ns;
WA <= '0'; WB <= '0'; WC <= '0'; WD <= '1'; wait for 10ns;
WA <= '0'; WB <= '0'; WC <= '0'; WD <= '0'; wait for 10ns;

wait;

end process;

end Behavioral;

```

Waveform/Results

The waveforms in figures 3 and 4 represent The SOP of the expression in the prelab and the POS circuit of the SOP to show the same outputs. In figure 3, the red line shows the voltage when both inputs are fluctuating between four combinations of 0s and 1s. Where green and yellow have the same inputs, as well as the white and red lines have the same input. The output is based on the green line, when it inputs a 1, the output is 1 and when the green line is 0, the output is 0.

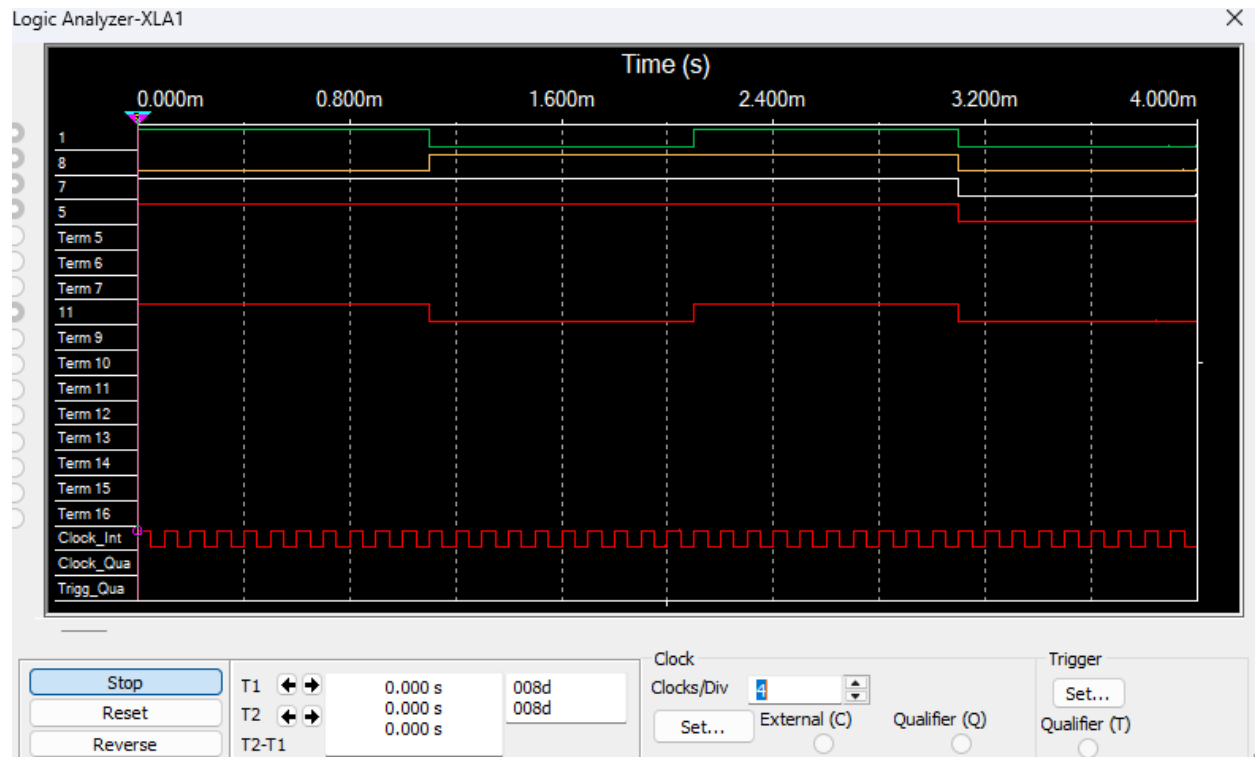


Figure 3: Waveform of the SOP equation of the prelab expression with four inputs

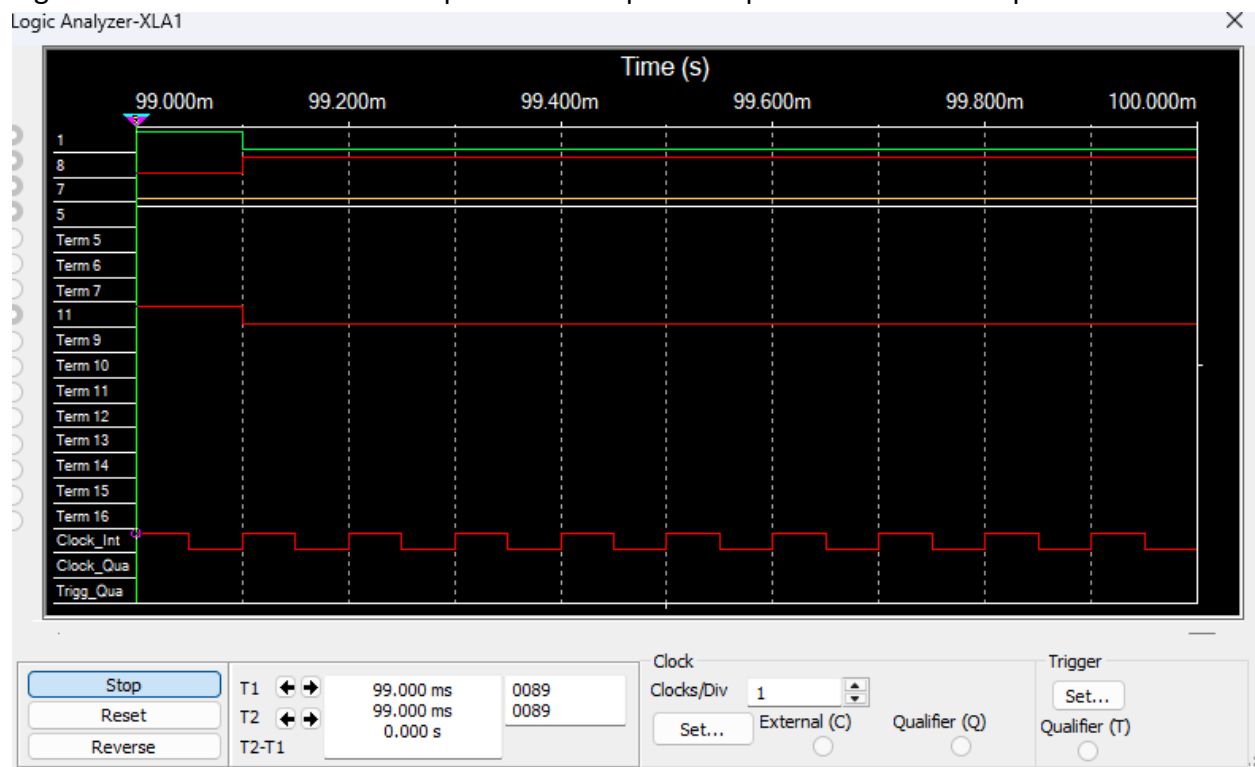


Figure 4: Waveform of the POS equation of the SOP circuit

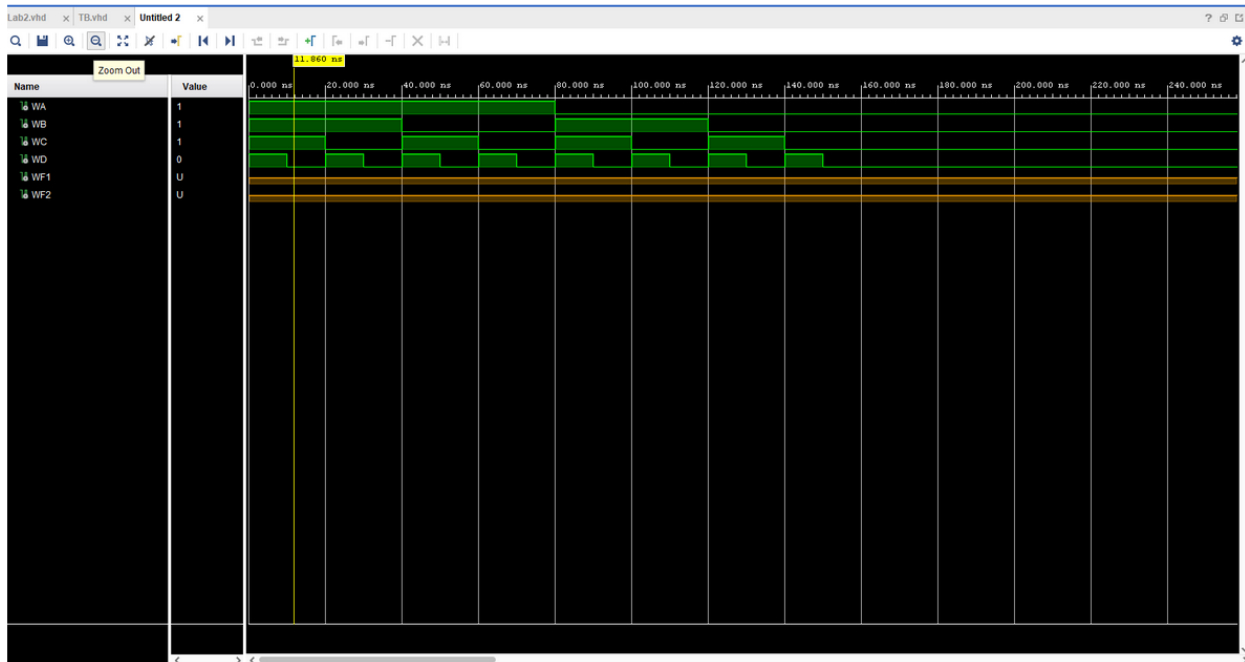


Figure 5: Behavioral simulation the SOP and POS expressions

The result in figure 6.1 shows the input that the program is submitting is 1100 and the output is 1 with the wiring being a SOP version of the prelab expression.

The result in figure 6.2 shows the input that the program is submitting is 1001 and the output is 1 with the wiring being a SOP version of the prelab expression.

The result in figure 6.3 shows the input that the program is submitting is 1011 and the output is 1 with the wiring being a SOP version of the prelab expression.

The result in figure 6.4 shows the input that the program is submitting is 1101 and the output is 1 with the wiring being a SOP version of the prelab expression.

The result in figure 6.5 shows the input that the program is submitting is 1111 and the output is 1 with the wiring being a SOP version of the prelab expression.

The result in figure 6.6 shows the input that the program is submitting is 0011 and the output is 1 with the wiring being a POS version of the SOP circuit.

The result in figure 6.7 shows the input that the program is submitting is 1001 and the output is 1 with the wiring being a POS version of the SOP circuit.

The result in figure 6.8 shows the input that the program is submitting is 1011 and the output is 1 with the wiring being a POS version of the SOP circuit.

The result in figure 6.9 shows the input that the program is submitting is 1101 and the output is 1 with the wiring being a POS version of the SOP circuit.

The result in figure 6.10 shows the input that the program is submitting is 1111 and the output is 1 with the wiring being a POS version of the SOP circuit.

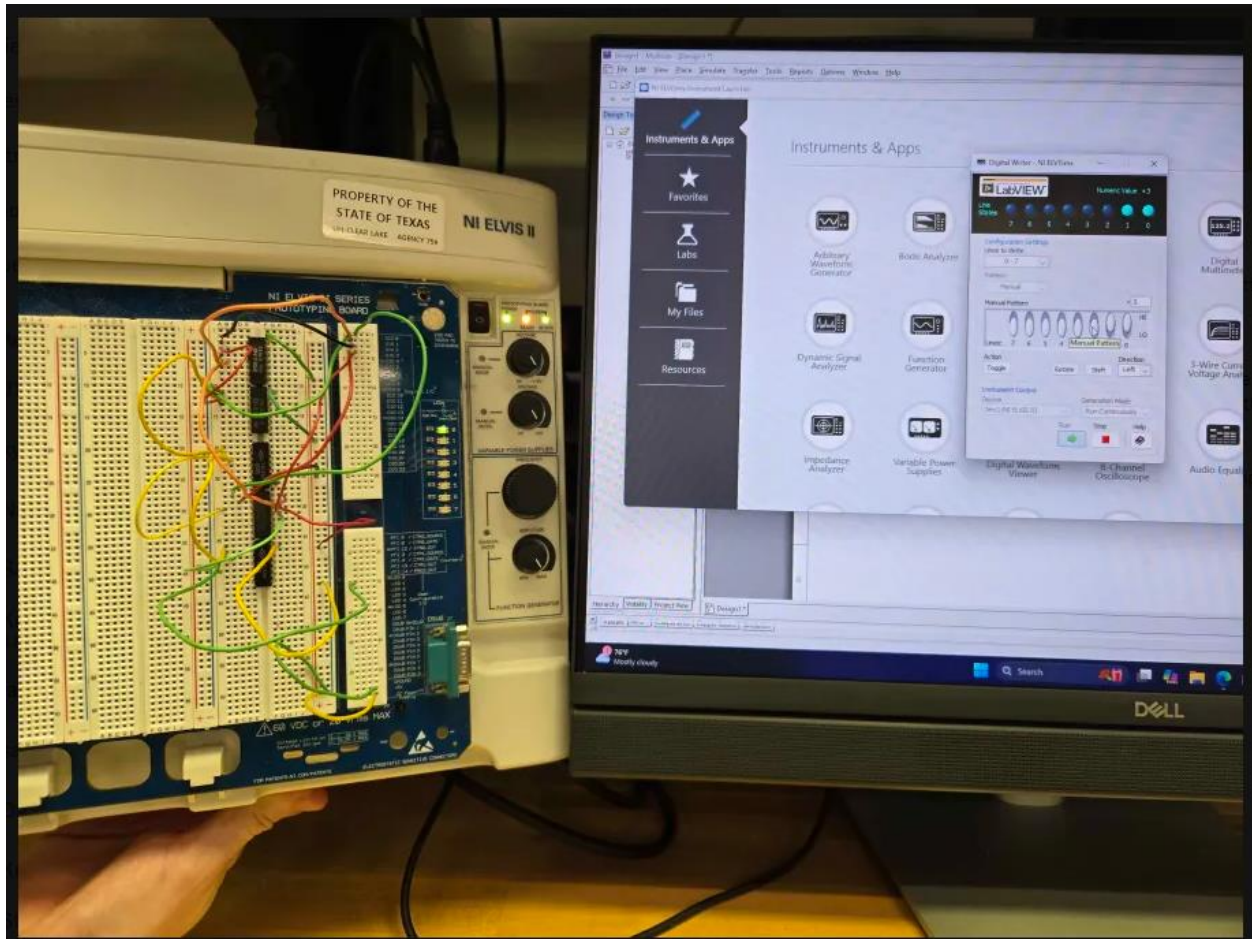


Figure 6.1: physical board showing the SOP with an input of the of the bits from LSB to MSB being 1100

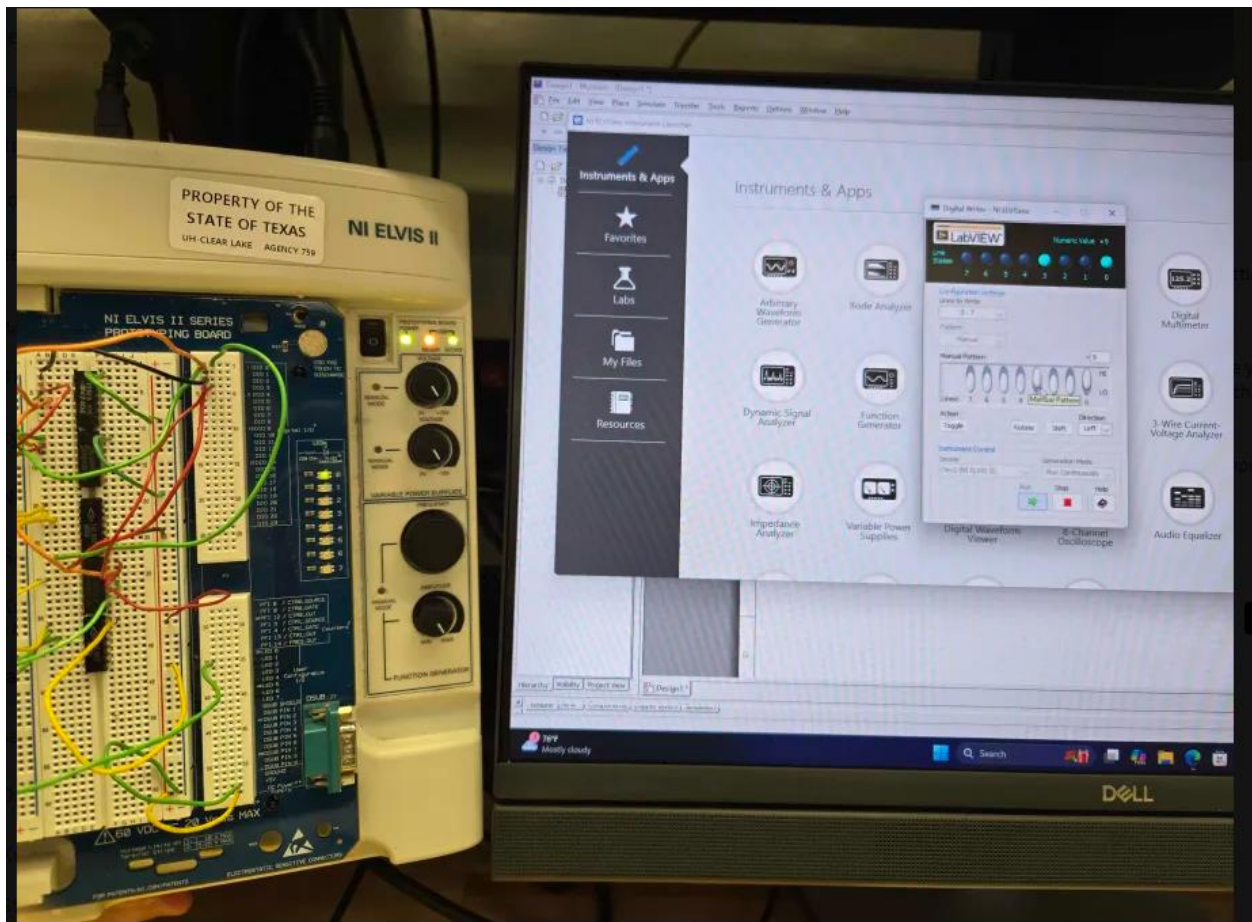


Figure 6.1: physical board showing the SOP with an input of the of the bits from LSB to MSB being 1001

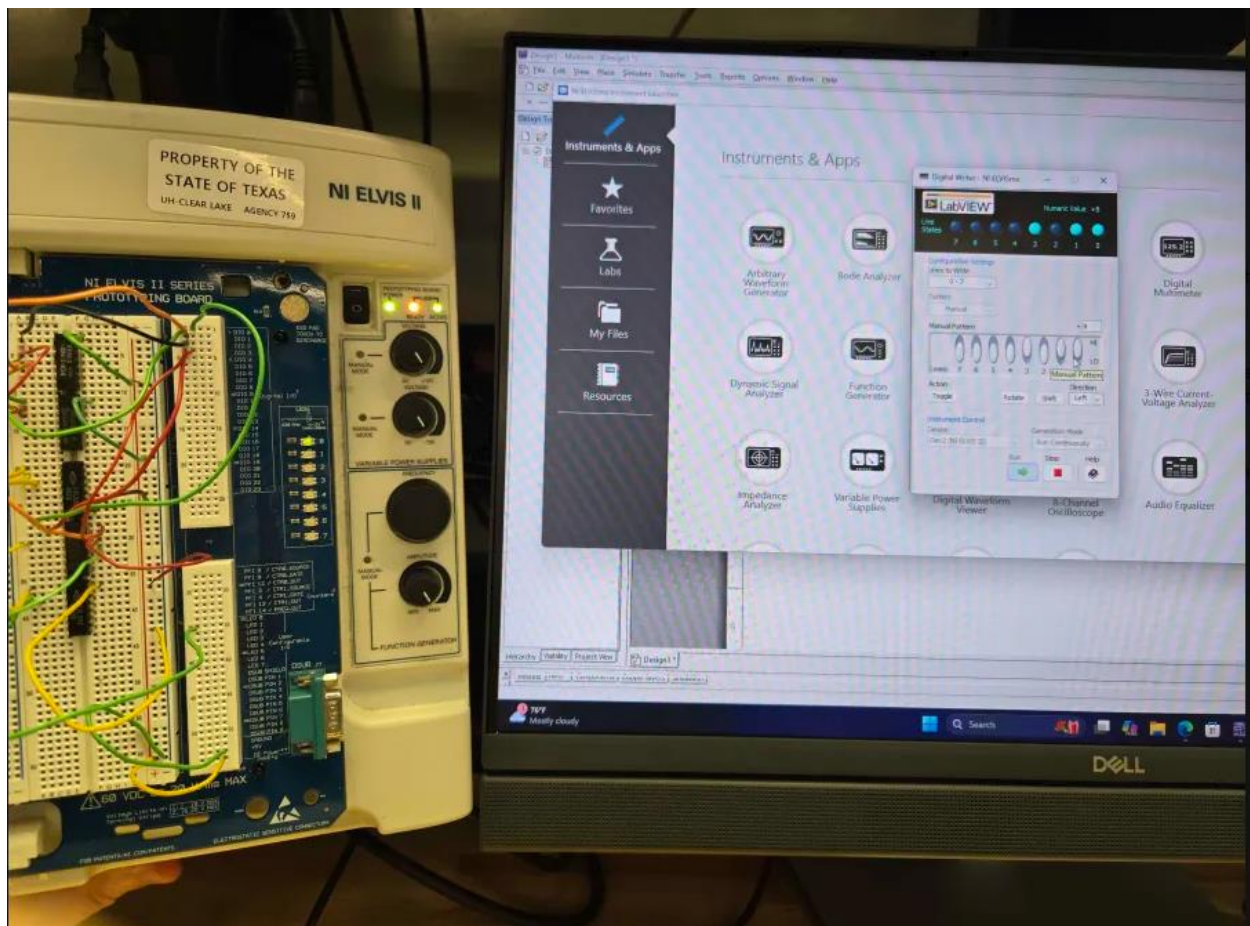


Figure 6.2: physical board showing the SOP with an input of the of the bits from LSB to MSB being 1011

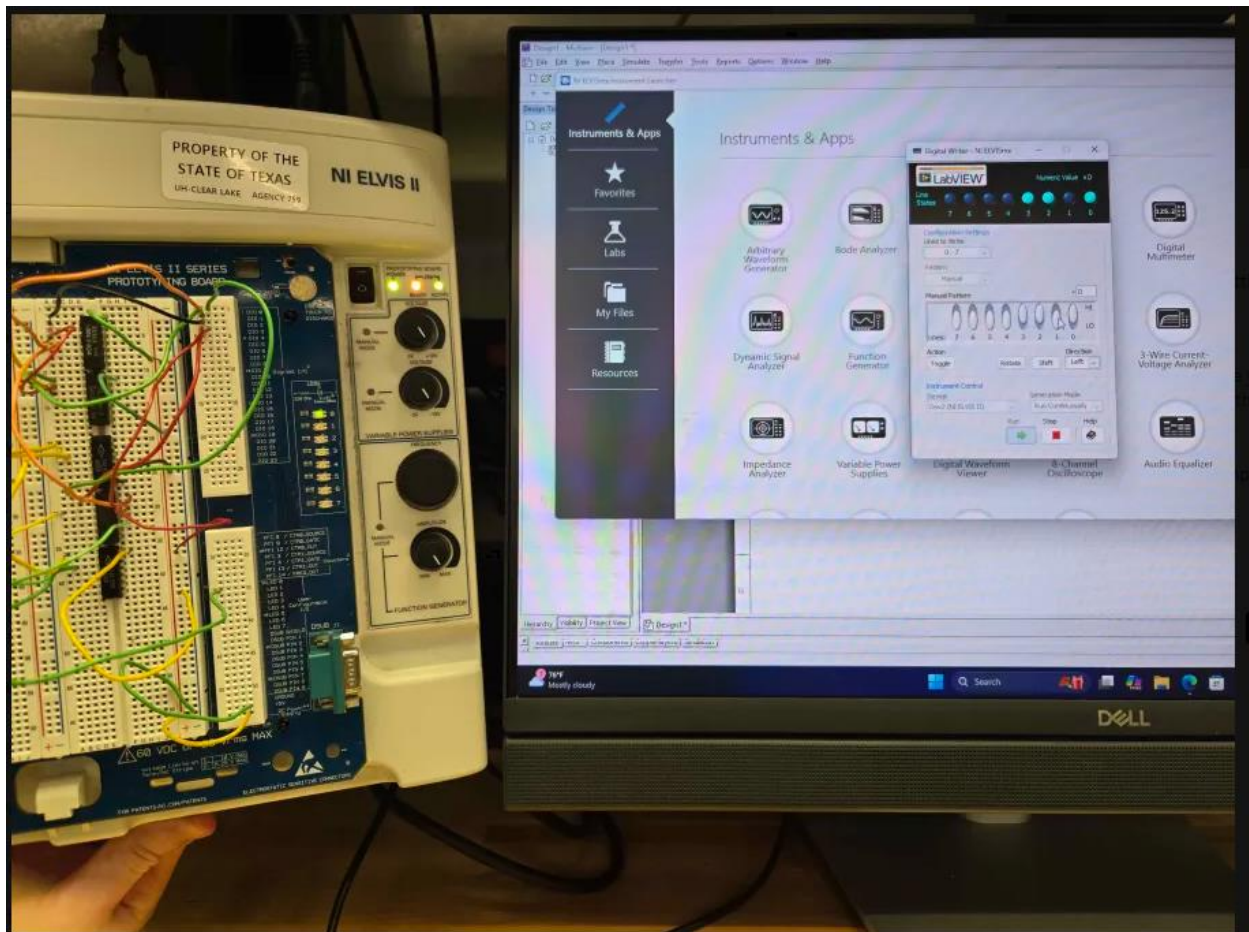


Figure 6.3: physical board showing the SOP with an input of the of the bits from LSB to MSB being 1101

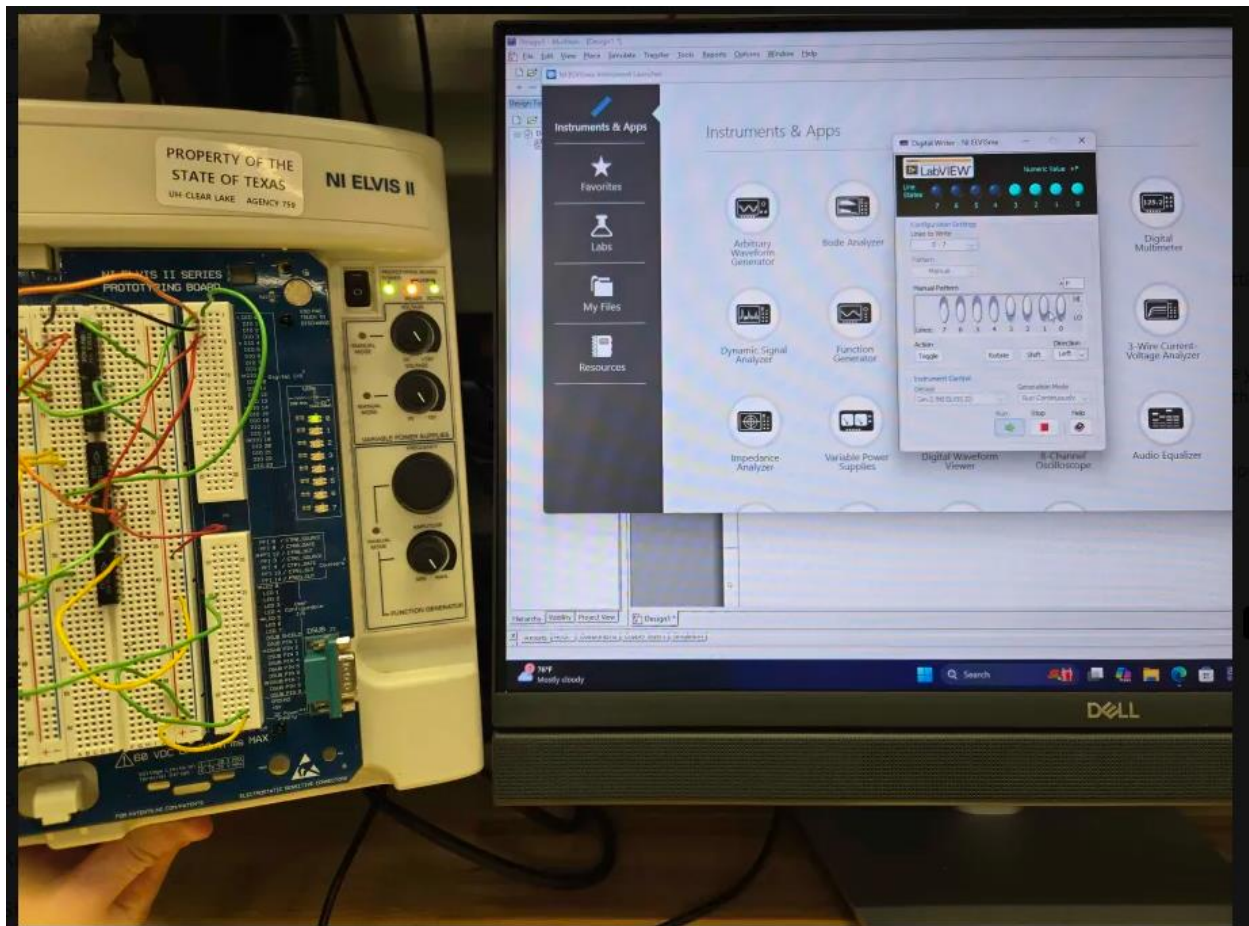


Figure 6.4: physical board showing the SOP with an input of the of the bits from LSB to MSB being 1111

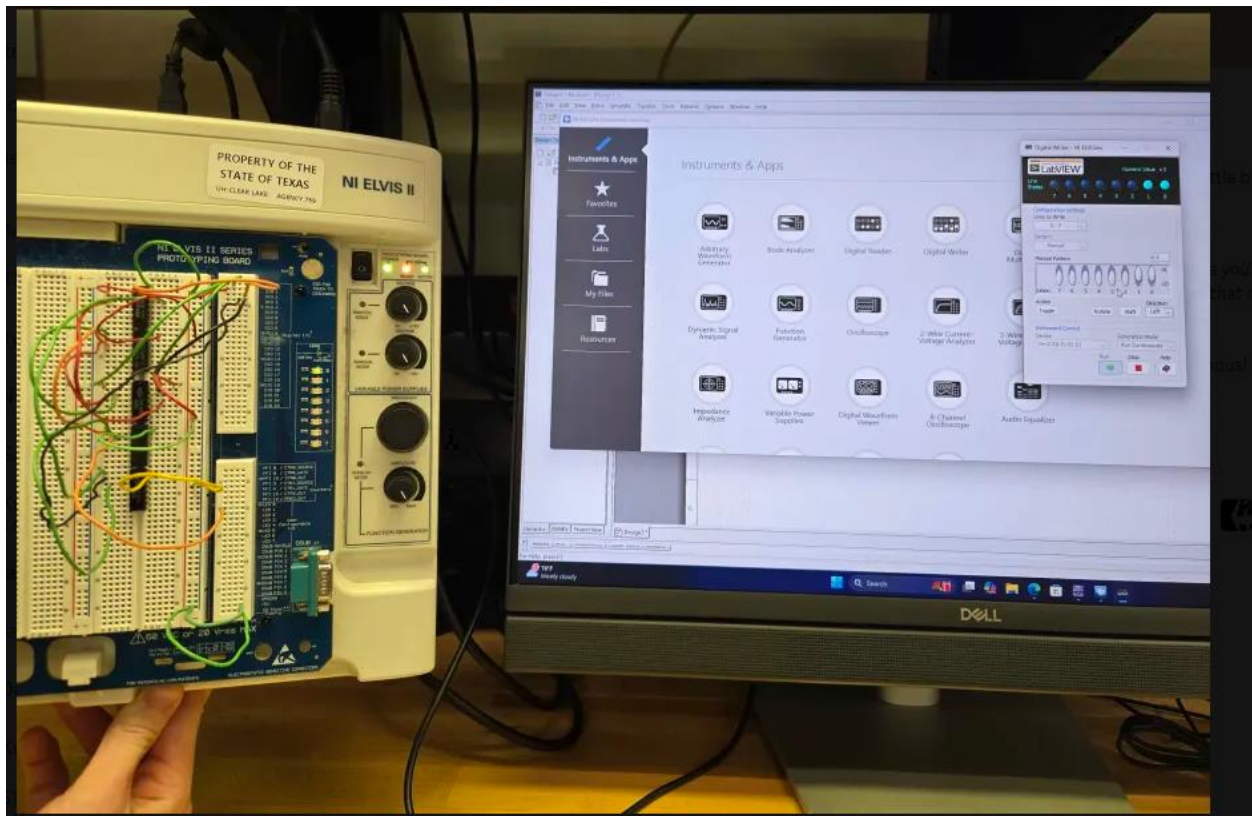


Figure 6.5: physical board showing the POS with an input of the of the bits from LSB to MSB being 0011

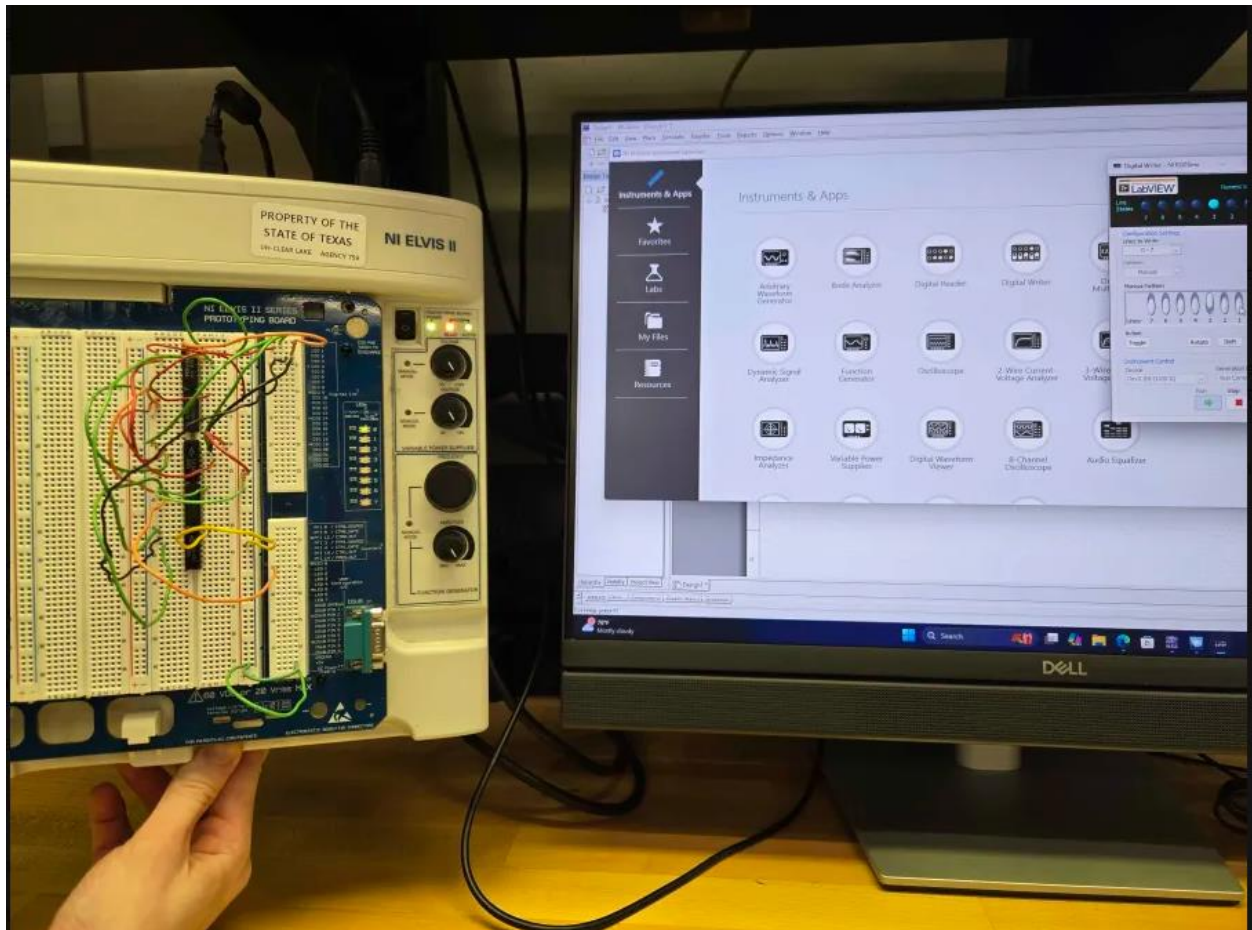


Figure 6.6: physical board showing the POS with an input of the of the bits from LSB to MSB being 0011

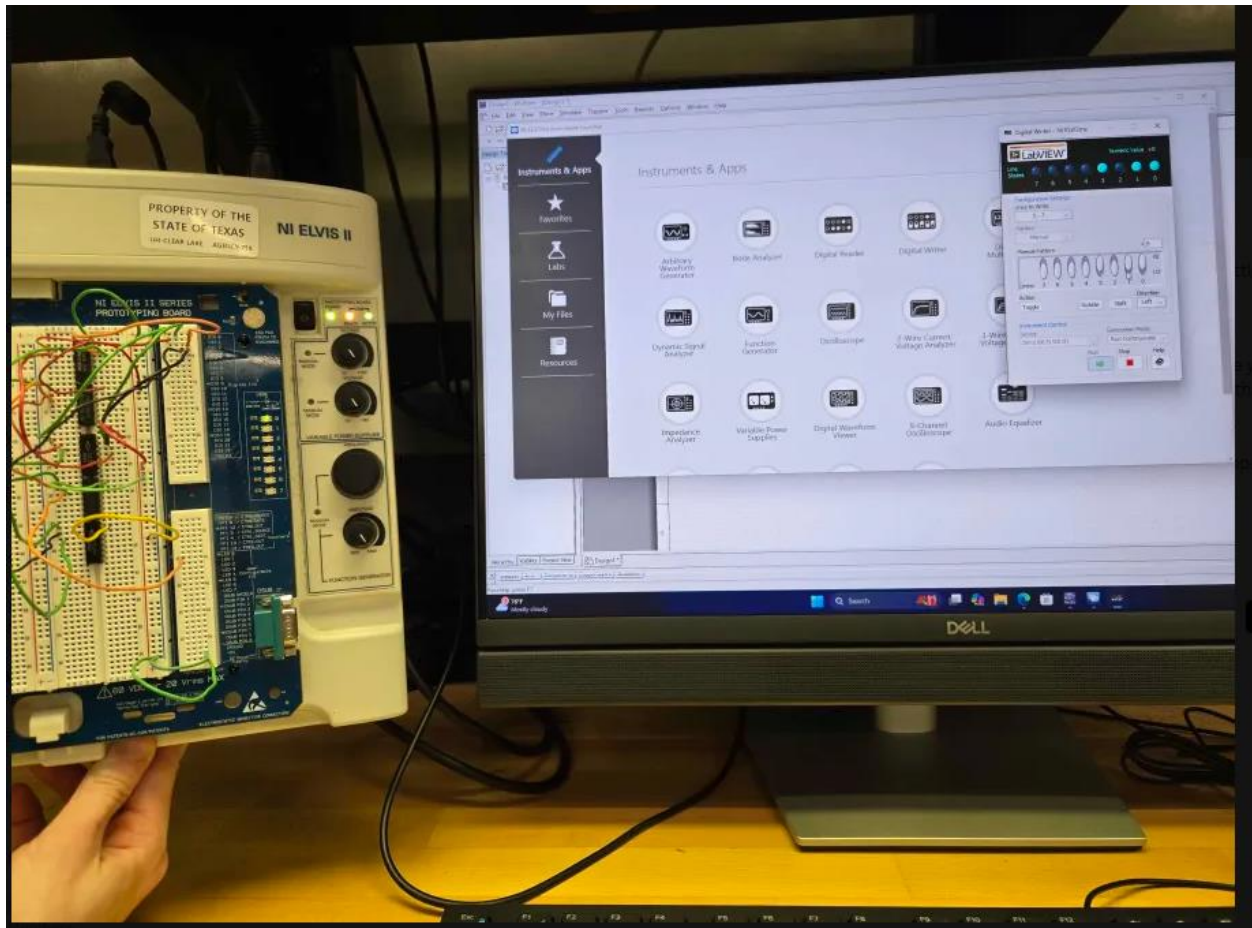


Figure 6.7: physical board showing the POS with an input of the of the bits from LSB to MSB being 1011

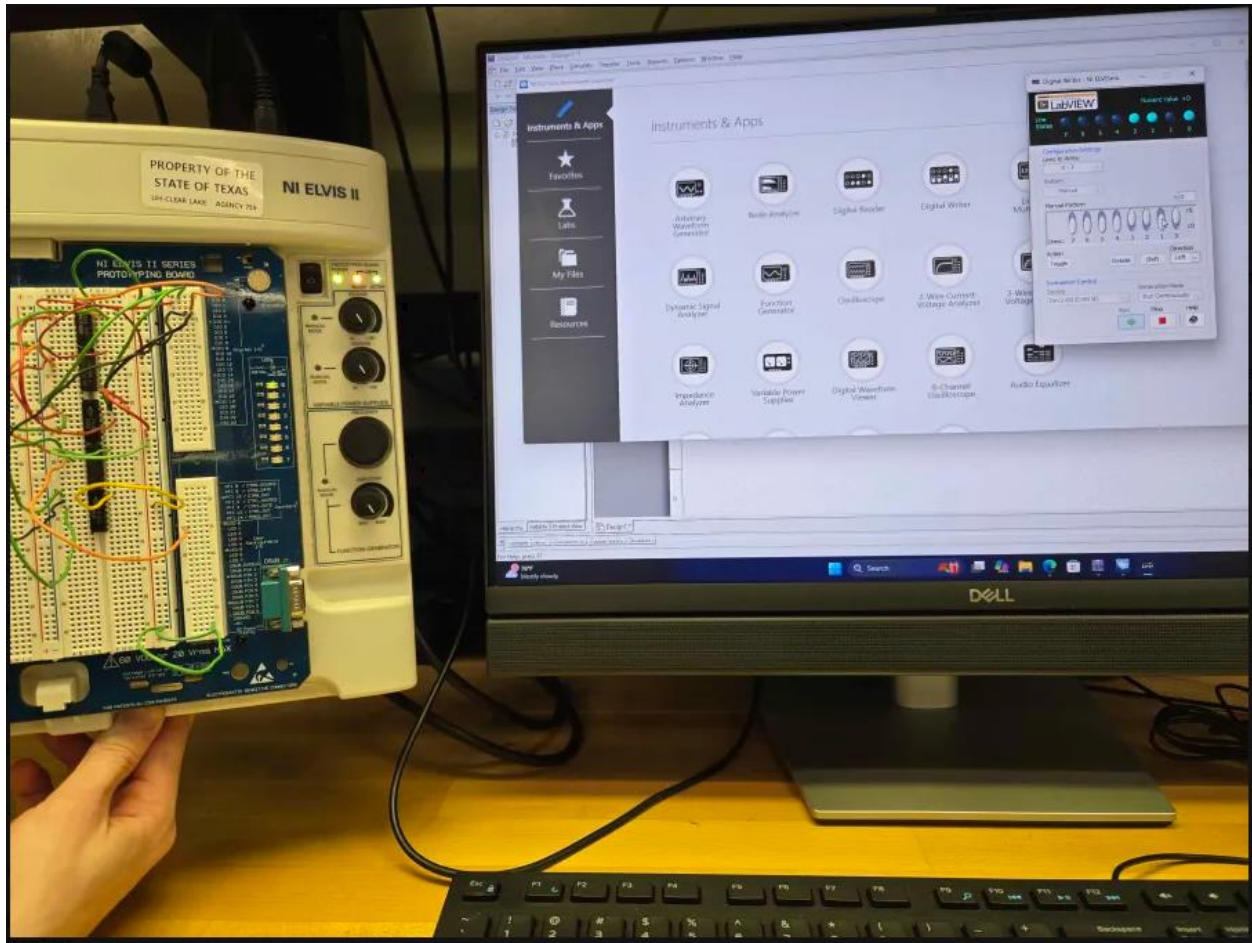


Figure 6.8: physical board showing the POS with an input of the of the bits from LSB to MSB

being 1101

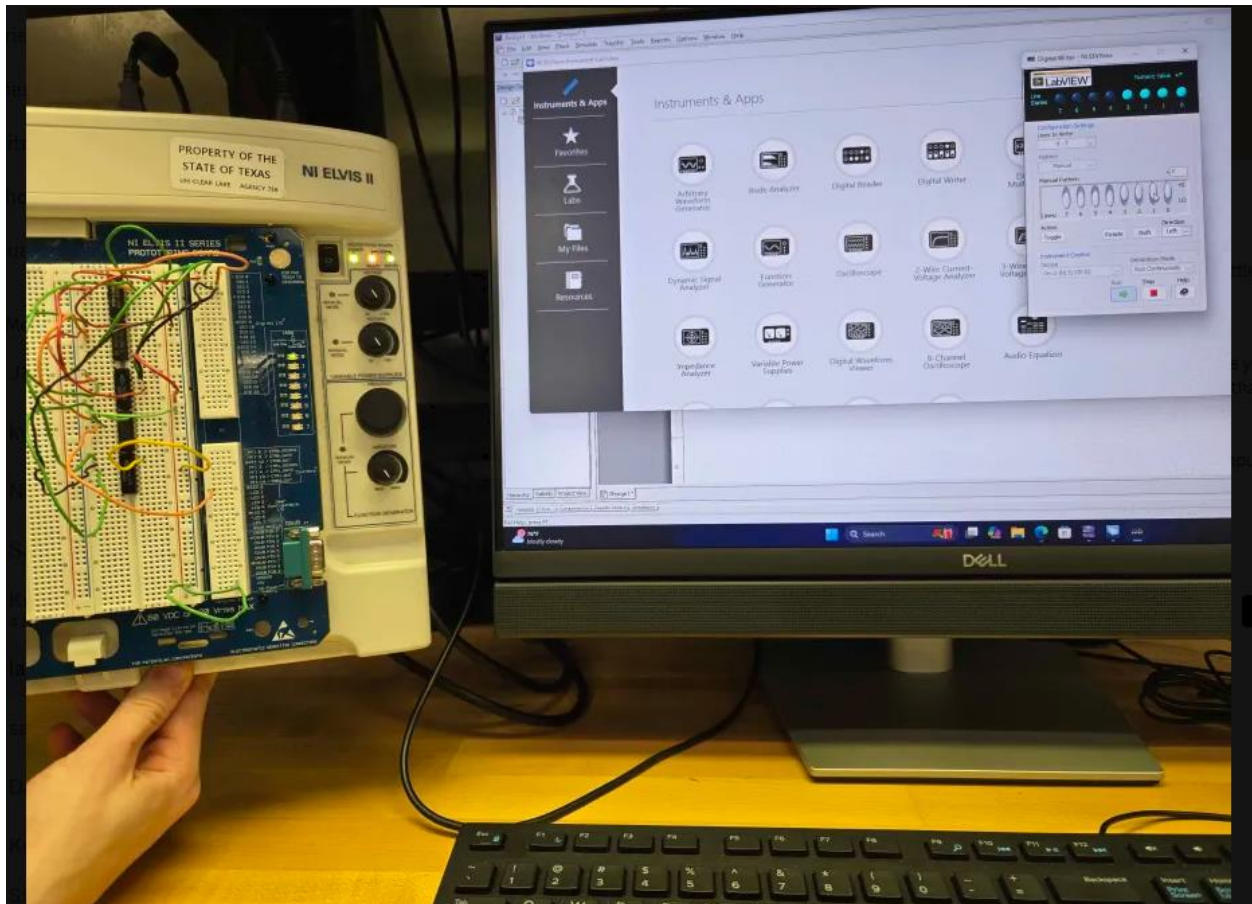


Figure 6.9: physical board showing the POS with an input of the of the bits from LSB to MSB being 1111

Conclusion

This project was designed to understand that the SOP and POS version of an expression can produce the same output given the same inputs. The main takeaway is that using Sum of products or product of sums can simplify a circuit and optimize the amount of gates needed to produce the same output as a longer, more complex circuit.